Retail Sales Data Analysis using Power BI

Objective

The objective of this project is to clean, model, and analyze retail sales data using Power BI. The project focuses on identifying the distribution of product categories across different countries and answering key business questions using data visualization.

Key Learnings

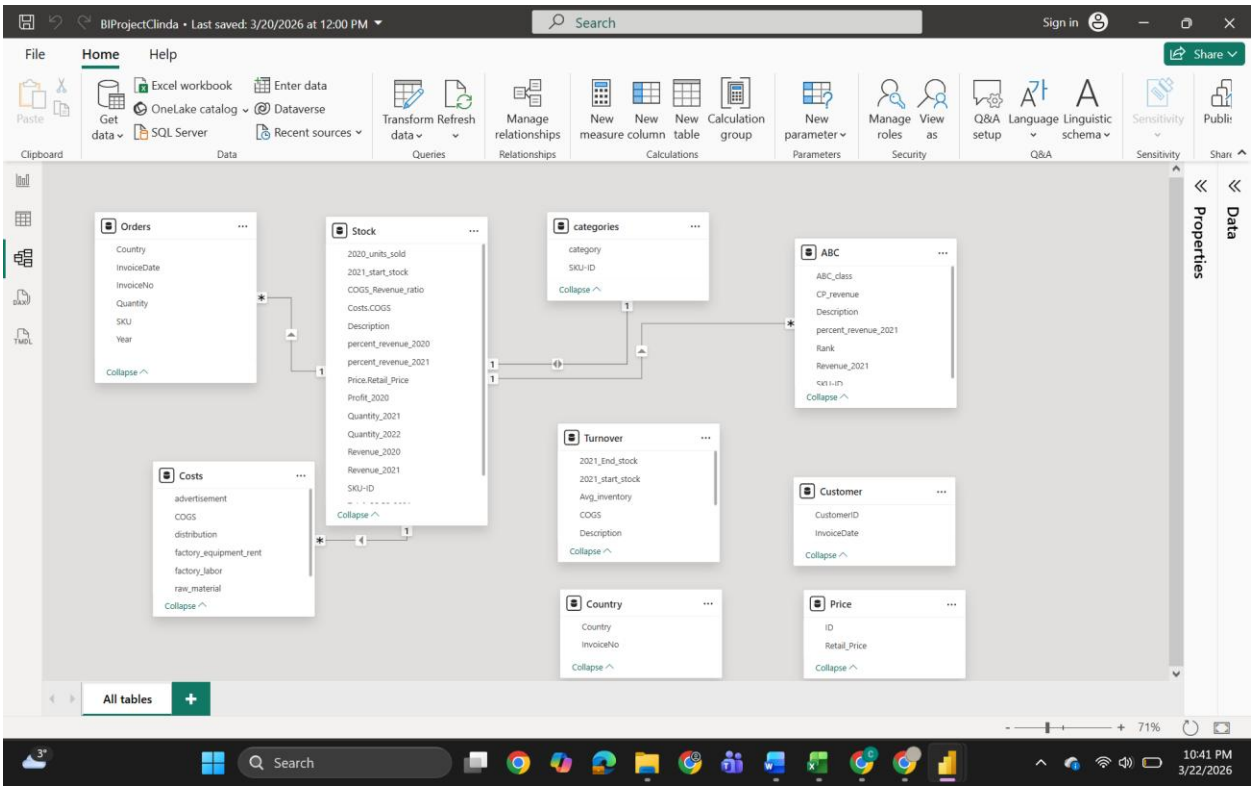
- Data modeling significantly impacts results
- Relationships can introduce duplication if not handled carefully
- Merging tables can simplify analysis
- Always validate results manually
- Visualization results must be cross-checked with raw data

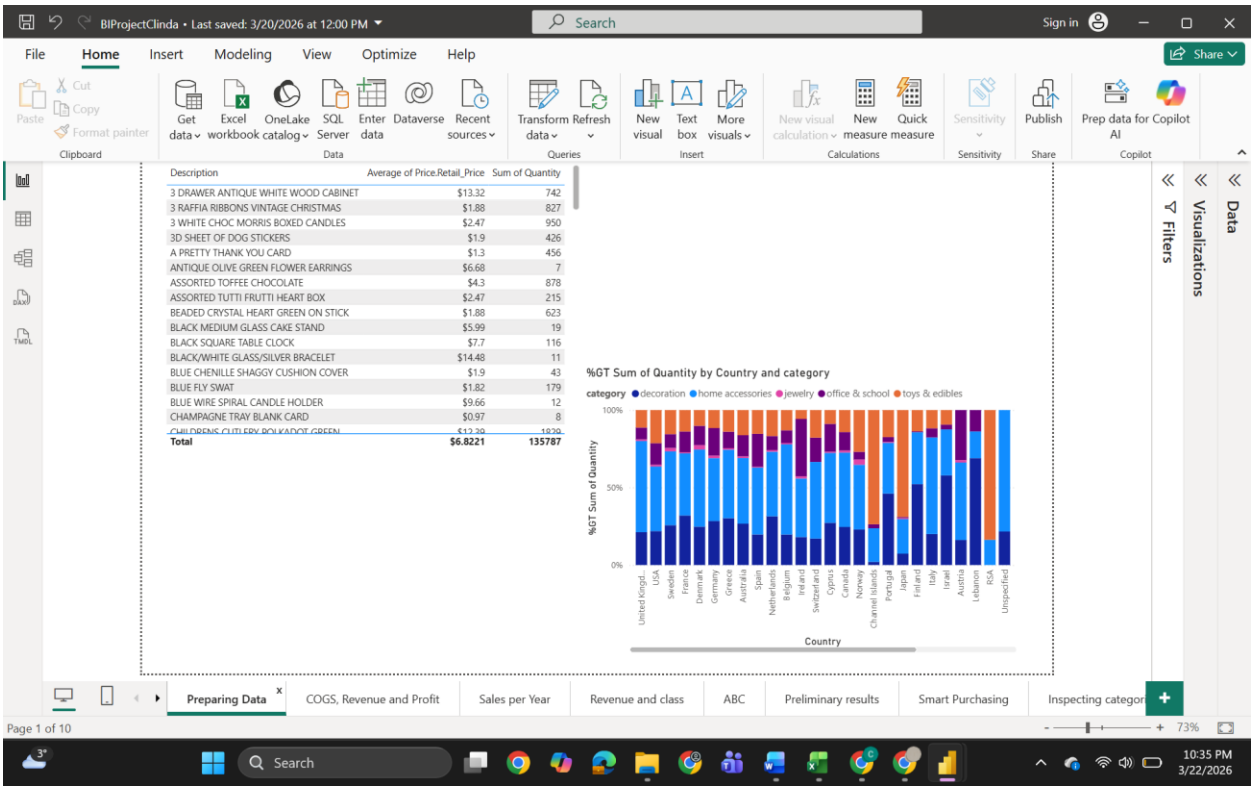
Issue Encountered

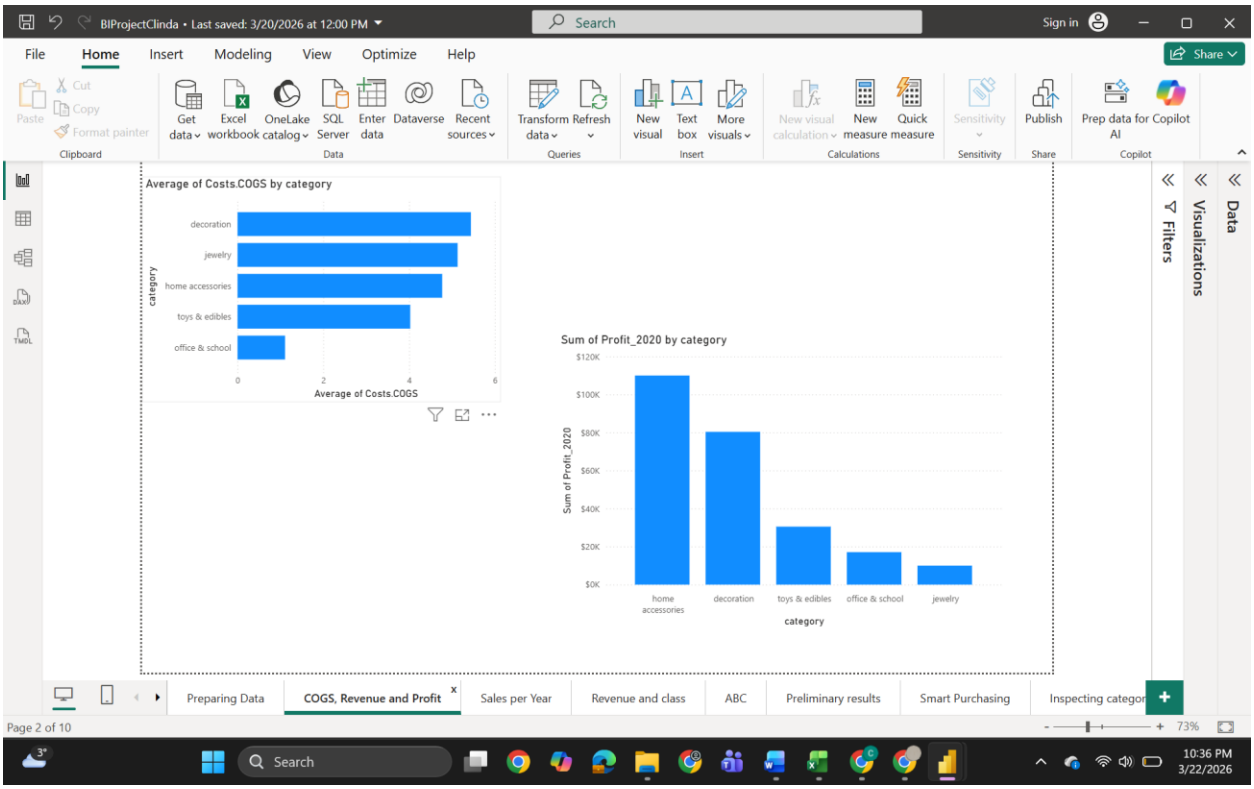
Using relationships initially caused:

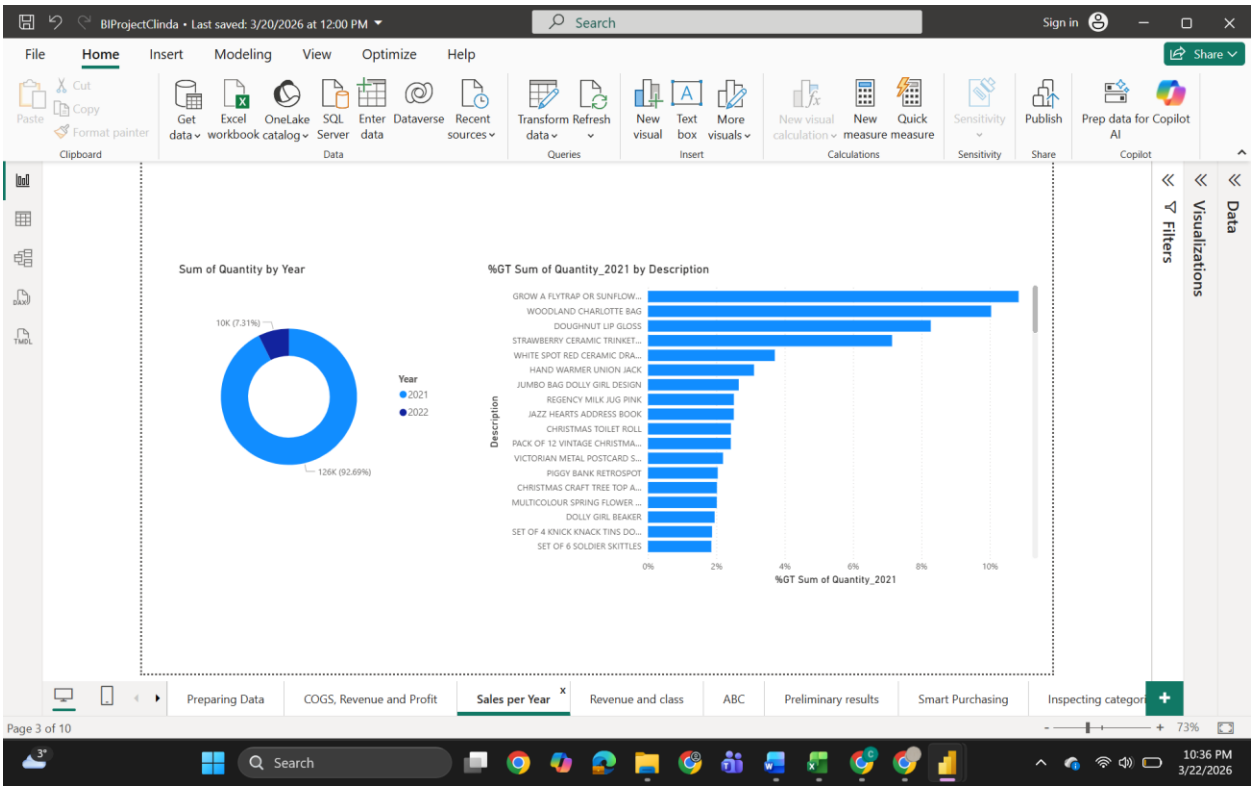
- Incorrect totals
- Data duplication
- Wrong percentage calculations
- Calculation errors with the table

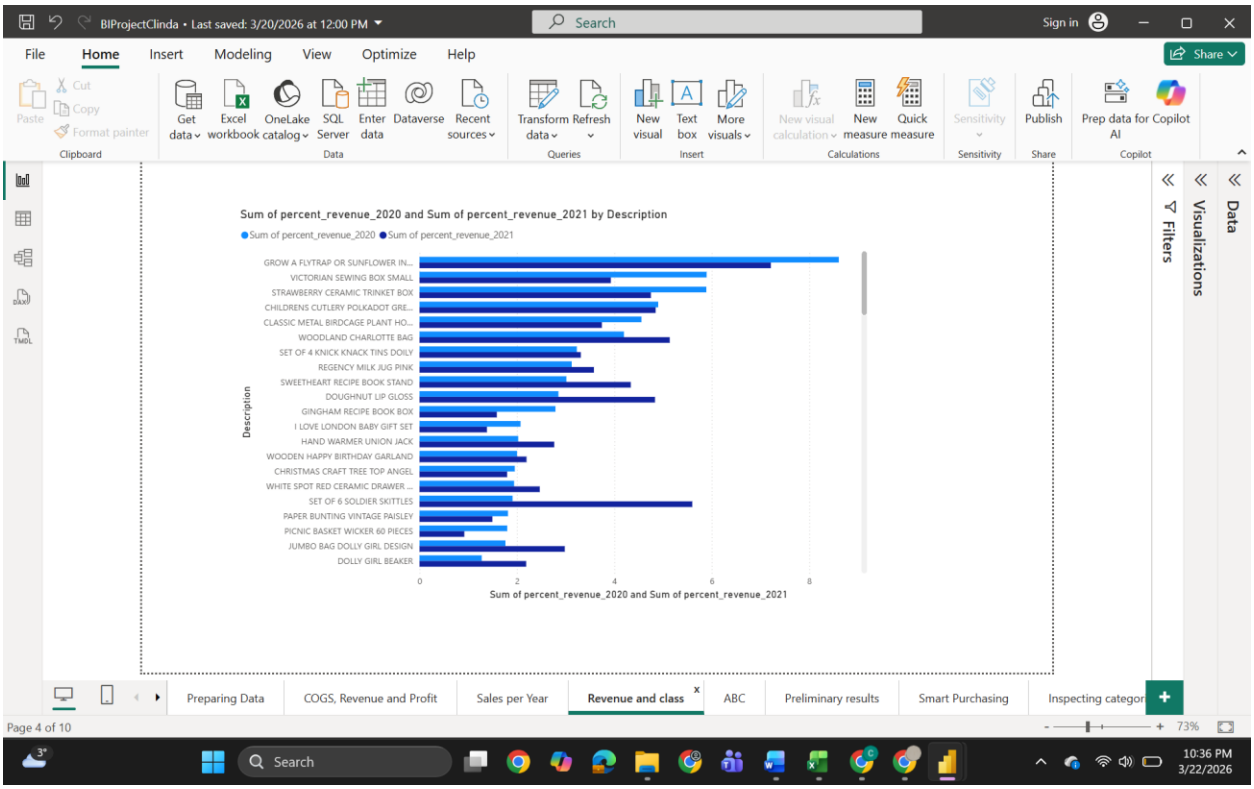
Screenshots

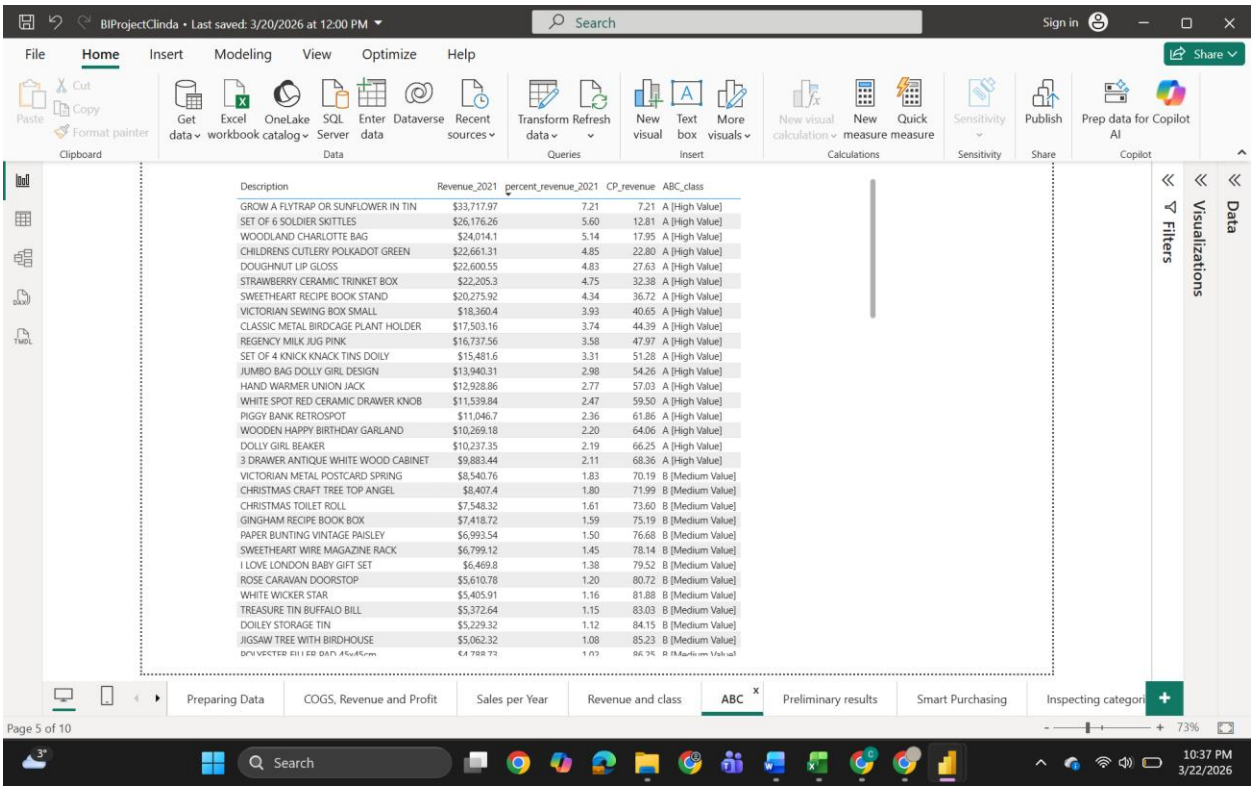




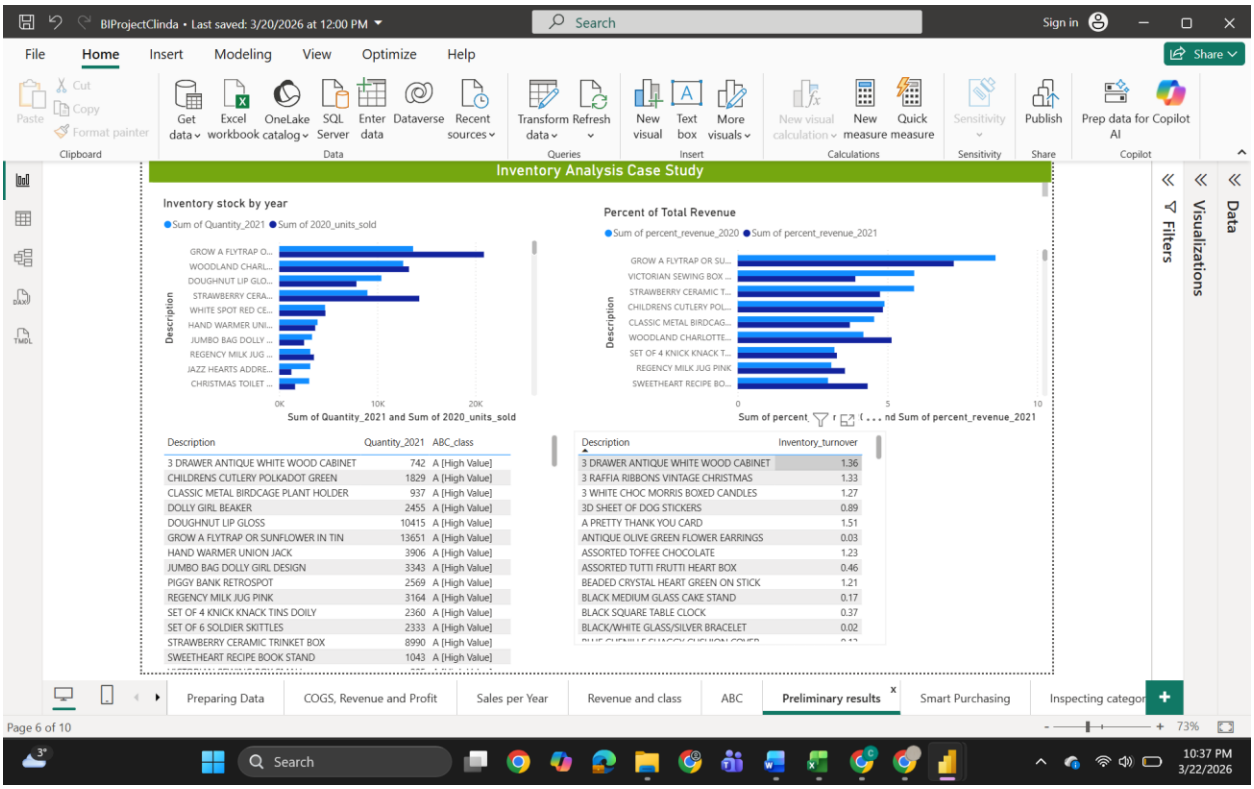

BIProjectClinda • Last saved: 3/20/2026 at 12:00 PM
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Average of Costs.COGS by category
decoration
jewelry
home accessories
toys & edibles
office & school
category
Average of Costs.COGS
Sum of Profit_2020 by category
$120K
$100K
$80K
$60K
$40K
$20K
$0K
Sum of Profit_2020
home accessories
decoration
toys & edibles
office & school
jewelry
category
Preparing Data
COGS, Revenue and Profit
Sales per Year
Revenue and class
ABC
Preliminary results
Smart Purchasing
Page 2 of 10
73%
10:36 PM
3/22/2026

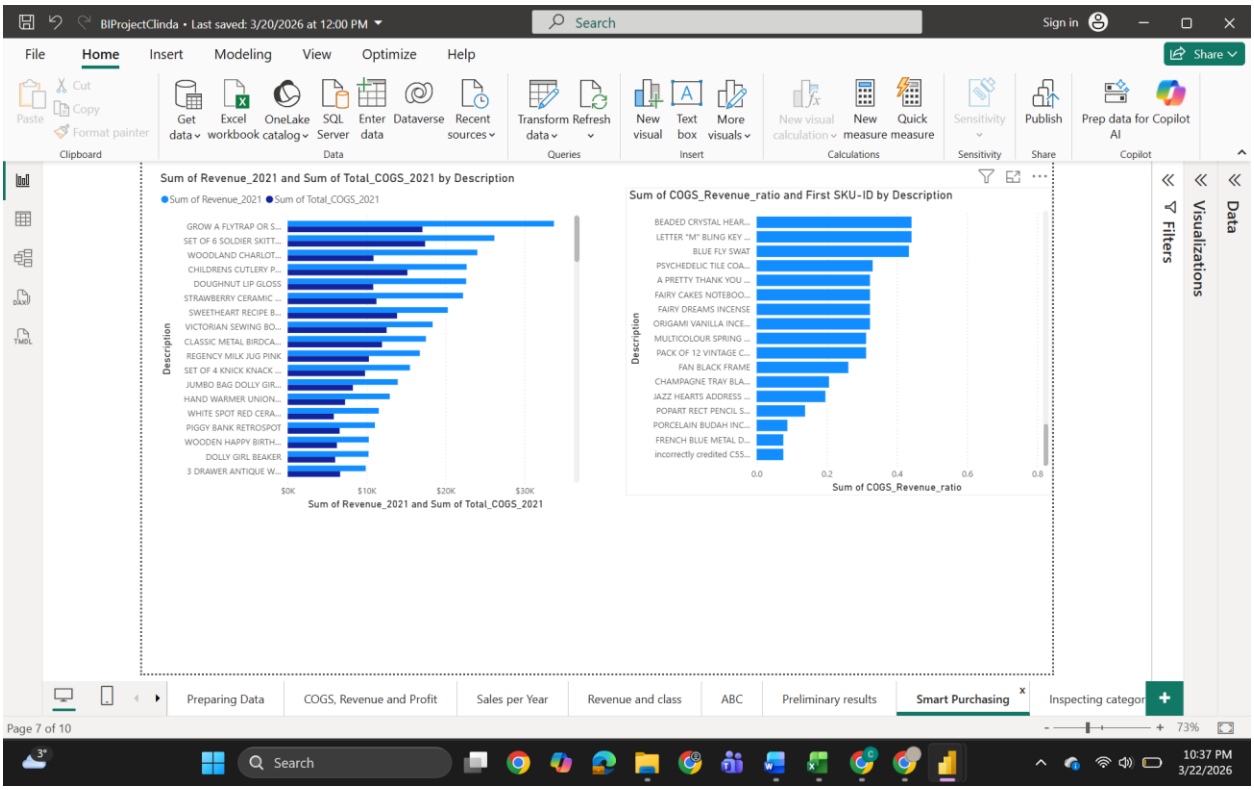

BIProjectClinda • Last saved: 3/20/2026 at 12:00 PM
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Sum of Quantity by Year
10K (7.31%)
126K (92.69%)
Year
2021
2022
%GT Sum of Quantity_2021 by Description
GROW A FLYTRAP OR SUNFLOW...
WOODLAND CHARLOTTE BAG
DOUGHNUT LIP GLOSS
STRAWBERRY CERAMIC TRINKET...
WHITE SPOT RED CERAMIC DRA...
HAND WARMER UNION JACK
JUMBO BAG DOLLY GIRL DESIGN
REGENCY MILK JUG PINK
JAZZ HEARTS ADDRESS BOOK
CHRISTMAS TOILET ROLL
PACK OF 12 VINTAGE CHRISTMA...
VICTORIAN METAL POSTCARD S...
PIGGY BANK RETROSPOT
CHRISTMAS CRAFT TREE TOP A...
MULTICOLOUR SPRING FLOWER ...
DOLLY GIRL BEAKER
SET OF 4 KNICK KNACK TINS DO...
SET OF 6 SOLDIER SKITTLES
Description
0%
2%
4%
6%
8%
10%
%GT Sum of Quantity_2021
Preparing Data
COGS, Revenue and Profit
Sales per Year
Revenue and class
ABC
Preliminary results
Smart Purchasing
Page 3 of 10
73%
10:36 PM
3/22/2026

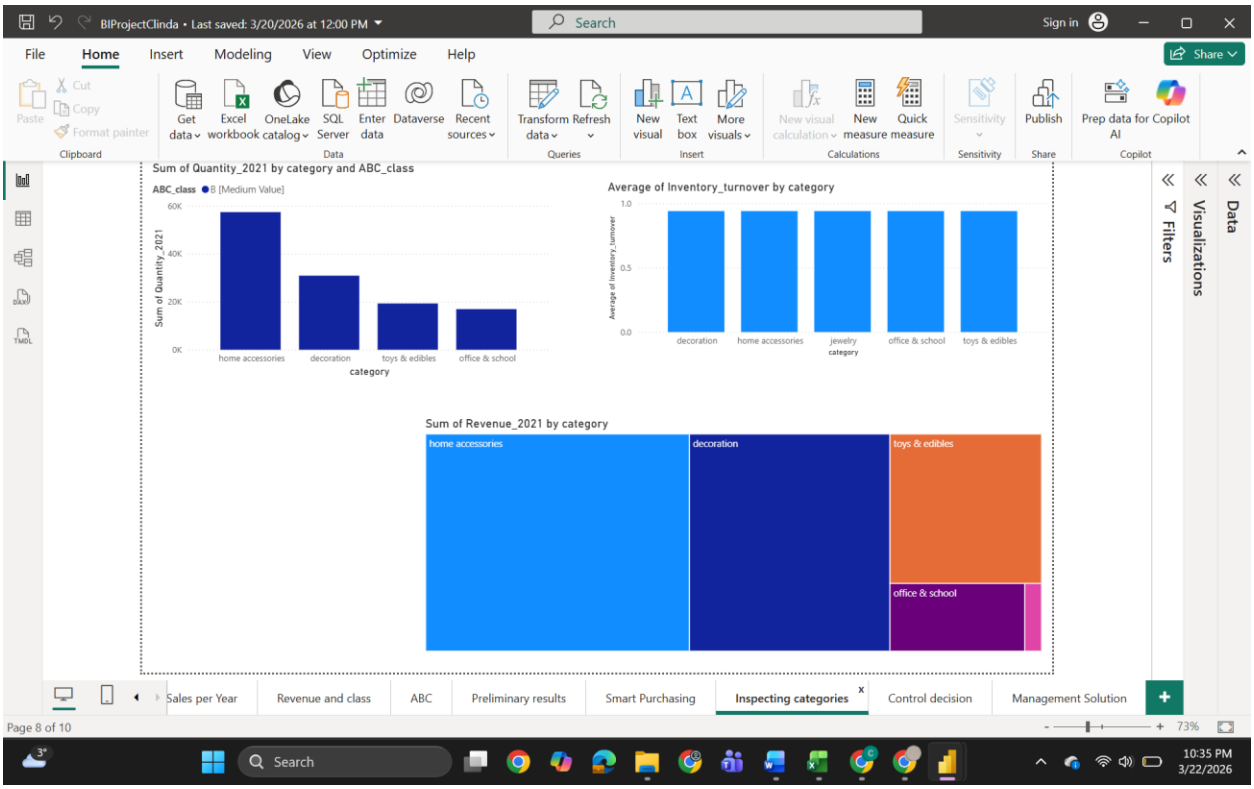


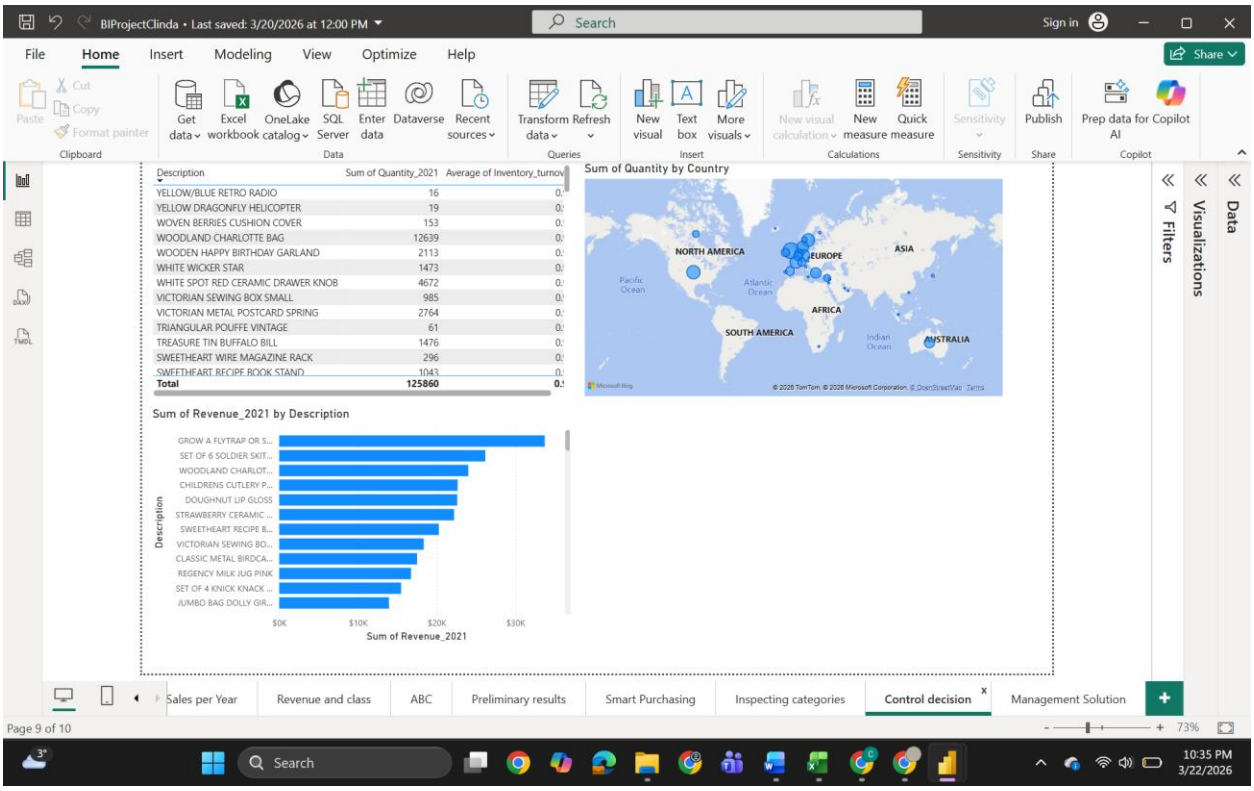


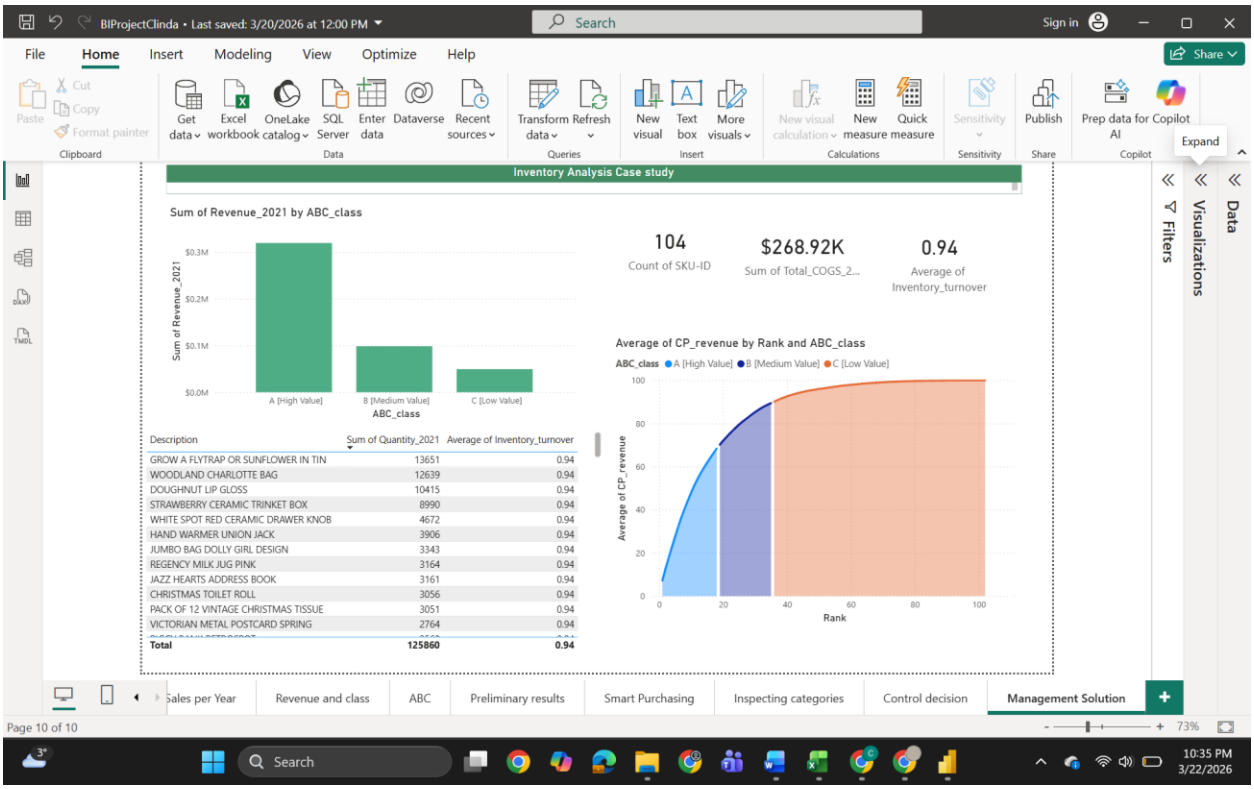
Description	Revenue_2021	percent_revenue_2021	CP_revenue	ABC_class
GROW A FLYTRAP OR SUNFLOWER IN TIN	$33,717.97	7.21	7.21	A [High Value]
SET OF 6 SOLDIER SKITTLES	$26,176.26	5.60	12.81	A [High Value]
WOODLAND CHARLOTTE BAG	$24,014.1	5.14	17.95	A [High Value]
CHILDRENS CUTLERY POLKADOT GREEN	$22,661.31	4.85	22.80	A [High Value]
DOUGHNUT LIP GLOSS	$22,600.55	4.83	27.63	A [High Value]
STRAWBERRY CERAMIC TRINKET BOX	$22,205.3	4.75	32.38	A [High Value]
SWEETHEART RECIPE BOOK STAND	$20,275.92	4.34	36.72	A [High Value]
VICTORIAN SEWING BOX SMALL	$18,360.4	3.93	40.65	A [High Value]
CLASSIC METAL BIRDCAGE PLANT HOLDER	$17,503.16	3.74	44.39	A [High Value]
REGENCY MILK JUG PINK	$16,737.56	3.58	47.97	A [High Value]
SET OF 4 KNICK KNACK TINS DOILY	$15,481.6	3.31	51.28	A [High Value]
JUMBO BAG DOLLY GIRL DESIGN	$13,940.31	2.98	54.26	A [High Value]
HAND WARMER UNION JACK	$12,928.86	2.77	57.03	A [High Value]
WHITE SPOT RED CERAMIC DRAWER KNOB	$11,539.84	2.47	59.50	A [High Value]
PIGGY BANK RETROSPOT	$11,046.7	2.36	61.86	A [High Value]
WOODEN HAPPY BIRTHDAY GARLAND	$10,269.18	2.20	64.06	A [High Value]
DOLLY GIRL BEAKER	$10,237.35	2.19	66.25	A [High Value]
3 DRAWER ANTIQUE WHITE WOOD CABINET	$9,883.44	2.11	68.36	A [High Value]
VICTORIAN METAL POSTCARD SPRING	$8,540.76	1.83	70.19	B [Medium Value]
CHRISTMAS CRAFT TREE TOP ANGEL	$8,407.4	1.80	71.99	B [Medium Value]
CHRISTMAS TOILET ROLL	$7,548.32	1.61	73.60	B [Medium Value]
GINGHAM RECIPE BOOK BOX	$7,418.72	1.59	75.19	B [Medium Value]
PAPER BUNTING VINTAGE PAISLEY	$6,993.54	1.50	76.68	B [Medium Value]
SWEETHEART WIRE MAGAZINE RACK	$6,799.12	1.45	78.14	B [Medium Value]
I LOVE LONDON BABY GIFT SET	$6,469.8	1.38	79.52	B [Medium Value]
ROSE CARAVAN DOORSTOP	$5,610.78	1.20	80.72	B [Medium Value]
WHITE WICKER STAR	$5,405.91	1.16	81.88	B [Medium Value]
TREASURE TIN BUFFALO BILL	$5,372.64	1.15	83.03	B [Medium Value]
DOILEY STORAGE TIN	$5,229.32	1.12	84.15	B [Medium Value]
JIGSAW TREE WITH BIRDHOUSE	$5,062.32	1.08	85.23	B [Medium Value]
POLYESTER FILLER PAD 45x45cm	$4,788.73	1.02	86.25	B [Medium Value]


BIProjectClinda • Last saved: 3/20/2026 at 12:00 PM
Inventory Analysis Case Study
Inventory stock by year
Sum of Quantity_2021
Sum of 2020_units_sold
Sum of Quantity_2021 and Sum of 2020_units_sold
Percent of Total Revenue
Sum of percent_revenue_2020
Sum of percent_revenue_2021
Description
Quantity_2021
ABC_class
3 DRAWER ANTIQUE WHITE WOOD CABINET 742 A [High Value]
CHILDRENS CUTLERY POLKADOT GREEN 1829 A [High Value]
CLASSIC METAL BIRDCAGE PLANT HOLDER 937 A [High Value]
DOLLY GIRL BEAKER 2455 A [High Value]
DOUGHNUT LIP GLOSS 10415 A [High Value]
GROW A FLYTRAP OR SUNFLOWER IN TIN 13651 A [High Value]
HAND WARMER UNION JACK 3906 A [High Value]
JUMBO BAG DOLLY GIRL DESIGN 3343 A [High Value]
PIGGY BANK RETROSPOT 2569 A [High Value]
REGENCY MILK JUG PINK 3164 A [High Value]
SET OF 4 KNICK KNACK TINS DOILY 2360 A [High Value]
SET OF 6 SOLDIER SKITTLES 2333 A [High Value]
STRAWBERRY CERAMIC TRINKET BOX 8990 A [High Value]
SWEETHEART RECIPE BOOK STAND 1043 A [High Value]
Description
Inventory_turnover
3 DRAWER ANTIQUE WHITE WOOD CABINET 1.36
3 RAFFIA RIBBONS VINTAGE CHRISTMAS 1.33
3 WHITE CHOC MORRIS BOXED CANDLES 1.27
3D SHEET OF DOG STICKERS 0.89
A PRETTY THANK YOU CARD 1.51
ANTIQUE OLIVE GREEN FLOWER EARRINGS 0.03
ASSORTED TOFFEE CHOCOLATE 1.23
ASSORTED TUTTI FRUTTI HEART BOX 0.46
BEADED CRYSTAL HEART GREEN ON STICK 1.21
BLACK MEDIUM GLASS CAKE STAND 0.17
BLACK SQUARE TABLE CLOCK 0.37
BLACK/WHITE GLASS/SILVER BRACELET 0.02
Preparing Data
COGS, Revenue and Profit
Sales per Year
Revenue and class
ABC
Preliminary results
Smart Purchasing
Page 6 of 10


BIProjectClinda • Last saved: 3/20/2026 at 12:00 PM
Sum of Revenue_2021 and Sum of Total_COGS_2021 by Description
Sum of Revenue_2021
Sum of Total_COGS_2021
Sum of Revenue_2021 and Sum of Total_COGS_2021
Sum of COGS_Revenue_ratio and First SKU-ID by Description
Sum of COGS_Revenue_ratio
Description
Preparing Data
COGS, Revenue and Profit
Sales per Year
Revenue and class
ABC
Preliminary results
Smart Purchasing
Page 7 of 10


BIProjectClinda • Last saved: 3/20/2026 at 12:00 PM
Sum of Quantity_2021 by category and ABC_class
ABC_class B [Medium Value]
Sum of Quantity_2021
60K
40K
20K
0K
home accessories
decoration
toys & edibles
office & school
category
Average of Inventory_turnover by category
Average of Inventory_turnover
1.0
0.5
0.0
decoration
home accessories
jewelry
office & school
toys & edibles
category
Sum of Revenue_2021 by category
home accessories
decoration
toys & edibles
office & school
Sales per Year
Revenue and class
ABC
Preliminary results
Smart Purchasing
Inspecting categories
Control decision
Management Solution
Page 8 of 10
73%
10:35 PM
3/22/2026


BIProjectClinda • Last saved: 3/20/2026 at 12:00 PM
Description
Sum of Quantity_2021
Average of Inventory_turnov
YELLOW/BLUE RETRO RADIO 16
YELLOW DRAGONFLY HELICOPTER 19
WOVEN BERRIES CUSHION COVER 153
WOODLAND CHARLOTTE BAG 12639
WOODEN HAPPY BIRTHDAY GARLAND 2113
WHITE WICKER STAR 1473
WHITE SPOT RED CERAMIC DRAWER KNOB 4672
VICTORIAN SEWING BOX SMALL 985
VICTORIAN METAL POSTCARD SPRING 2764
TRIANGULAR POUFFE VINTAGE 61
TREASURE TIN BUFFALO BILL 1476
SWEETHEART WIRE MAGAZINE RACK 296
SWEETHEART RECIPE BOOK STAND 1043
Total 125860
Sum of Quantity by Country
NORTH AMERICA
EUROPE
ASIA
AFRICA
SOUTH AMERICA
AUSTRALIA
Pacific Ocean
Atlantic Ocean
Indian Ocean
Sum of Revenue_2021 by Description
GROW A FLYTRAP OR S...
SET OF 6 SOLDIER SKIT...
WOODLAND CHARLOT...
CHILDRENS CUTLERY P...
DOUGHNUT LIP GLOSS
STRAWBERRY CERAMIC ...
SWEETHEART RECIPE B...
VICTORIAN SEWING BO...
CLASSIC METAL BIRDCA...
REGENCY MILK JUG PINK
SET OF 4 KNICK KNACK ...
JUMBO BAG DOLLY GIR...
Description
$0K
$10K
$20K
$30K
Sum of Revenue_2021
Sales per Year
Revenue and class
ABC
Preliminary results
Smart Purchasing
Inspecting categories
Control decision
Management Solution
Page 9 of 10
73%
10:35 PM
3/22/2026



Description	Sum of Quantity_2021	Average of Inventory_turnover
GROW A FLYTRAP OR SUNFLOWER IN TIN	13651	0.94
WOODLAND CHARLOTTE BAG	12639	0.94
DOUGHNUT LIP GLOSS	10415	0.94
STRAWBERRY CERAMIC TRINKET BOX	8990	0.94
WHITE SPOT RED CERAMIC DRAWER KNOB	4672	0.94
HAND WARMER UNION JACK	3906	0.94
JUMBO BAG DOLLY GIRL DESIGN	3343	0.94
REGENCY MILK JUG PINK	3164	0.94
JAZZ HEARTS ADDRESS BOOK	3161	0.94
CHRISTMAS TOILET ROLL	3056	0.94
PACK OF 12 VINTAGE CHRISTMAS TISSUE	3051	0.94
VICTORIAN METAL POSTCARD SPRING	2764	0.94
Total	**125860**	**0.94**